
Vegetation Control Treatments to Favor Naturally Regenerated *Betula alleghaniensis* Saplings Following Seed-Tree Cut: Sapling Monitoring Two Years after Treatment

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Abstract

Control of competing vegetation is recommended to ensure successful Yellow birch (*Betula alleghaniensis* Britton) regeneration within juvenile stands that do not sustain high enough sapling densities of this species. Four contrasting vegetation control treatments were tested to determine their effect on the growth and vigor of eight-year-old *B. alleghaniensis* saplings regenerating after final cutting of a shelterwood seed cut. Vegetation control treatments were TC (total circular removal), PC (circular removal of codominant competing vegetation), TS (total semicircular removal on 180° section), and NC (no vegetation control). Two years after treatment application, diameter growth significantly improved in response to vegetation control treatments, whereas sapling height growth did not. This pattern of biomass allocation was directly related to sapling etiolation, which increased with decreasing severity of vegetation removal. As a result, application of vegetation control, especially TC and PC

treatments, was valuable in reducing signs of stress in saplings. However, increasing the severity of vegetation removal also made saplings more conspicuous to herbivores, which increased browsing, especially in the TC and PC treatments. Browsing was sufficient in some plots of the TC and PC treatments to overcome the vigor and diameter growth enhancements observed when browsing was negligible. In contrast to the TC and PC treatments, the TS treatment kept browsing very low while largely removing competition. The results suggest that *B. alleghaniensis* saplings established after final cutting of a shelterwood seed cut do take advantage of vegetation control treatments, but the decision to apply these treatments must include consideration of local herbivore population densities.

Key words: deer browsing, etiolation, saplings, vegetation control, Yellow birch.

Introduction

Among economically important hardwood species, Yellow birch (*Betula alleghaniensis*, Britton) is one of the most valuable found in the forests of northeastern North America. However, since the beginning of twentieth century, reports indicate a proportional decrease of *B. alleghaniensis* in favor of more shade-tolerant species such as *Acer saccharum*, Marsh. (Nolet et al. 2001b; Bouffard et al. 2003). Establishment and early survival are considered the most important factors limiting regeneration of *B. alleghaniensis* (Erdmann 1990). Poor establishment may be the result of diminishing seed sources caused by a long history of selective cuttings of vigorous seed producers

(Nolet et al. 2001a) and the general practice of winter harvesting, which avoids the soil disturbances needed for germination of this species (Houle 1992). Moreover, low survival and growth rates of *B. alleghaniensis* have been attributed to the diminishing occurrence of large canopy openings, as compared to previous decades, because of better fire control (Woods 2000). The extensive use of partial cutting (MRNQ 1998), which results in low light levels in forest understorey, has already been associated with the lack of *B. alleghaniensis* regeneration success (McClure et al. 2000). One opportunity to manage *B. alleghaniensis* in Canada is the use of the seed-tree method in which a large proportion of trees are removed (i.e., the creation of large openings), leaving a small number of seed-bearing trees of a desirable species. The uncut trees are anticipated to be the main source of seed for establishment of natural regeneration and are generally harvested in a final cutting.

Within juvenile stands with low *B. alleghaniensis* sapling densities, control of competing vegetation has

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also been recommended to ensure successful regeneration and sufficient crop tree establishment (Nolet et al. 2001a). Vegetation control treatments decrease mortality rates, increase productivity, and modify future stand composition (Godman & Marquis 1969; Roberge 1988; Heitzman & Nyland 1991; Miller 2000). To efficiently overcome *B. alleghaniensis* establishment deficiency and its low competitive ability versus more shade-intolerant and opportunistic species, vegetation control must be applied early in stand initiation (Godman & Marquis 1969; Hannah 1991).

Precautions must be taken before extensive application of vegetation control treatments. Several studies reported that substantial clearing of competing vegetation might affect stem quality by (1) delaying the natural process of lower branch pruning (e.g., Erdmann et al. 1981; Miller 2000), (2) creating new lateral branches (Godman & Marquis 1969), (3) increasing individual susceptibility to ice damage (Groninger et al. 2000), and (4) encouraging the main stem to fork (Godman & Marquis 1969; Erdmann et al. 1981). Furthermore, releasing stems and branches from surrounding competing vegetation increases their conspicuousness to herbivores (e.g., insects, rodents, cervidae), and the more severe the vegetation removal, the more damaging the predation (Kohn & Mooty 1971; Drolet 1978; Monthey 1984).

Four contrasting vegetation control treatments were thus initiated after final cutting using a seed-tree method in *B. alleghaniensis*-dominated stands to determine if follow-up vegetation treatment is required to secure desirable natural regeneration. We were particularly interested in (1) how varying intensities of vegetation removal treatments affected the growth of established *B. alleghaniensis* saplings two years after treatment application and (2) the importance of deer browsing on sapling growth, with respect to various intensities of vegetation removal.

Methods

Study Area

The study site was located in the south part of the Laverdière Lake landscape unit (Robitaille & Saucier 1998), around 60 km north of the city of Saint-Michel-des-Saints (Québec, Canada). This area belongs to the Balsam fir (*Abies balsamea* L.)–*Betula alleghaniensis* bioclimatic domain, but Sugar maple (*Acer saccharum*, Marsh.) is also codominant where soils are well drained. The area is characterized by hills of moderate height (mean elevation 520 m) with gentle slopes. Soils are composed of a deep till (0.5–1 m) topped by a sandy podzolic layer containing high rock densities (Robitaille & Saucier 1998). Mean annual temperature is 2.5°C, and annual precipitation ranges from 1,000 to 1,100 mm, with around 300 mm as snow (Robitaille & Saucier 1998).

Experimental Design

During September 2001, 32 square plots, each 0.25 ha in size, were established in an area harvested in 1994 using a seed-tree method with a final cut in 2000. Initial stand composition was dominated by *B. alleghaniensis*, and mean stand age prior to harvest was about 90 years. Plots were distributed in order to place each plot center at least at 75 m from surrounding plot centers.

In June 2002, 20 *B. alleghaniensis* codominant saplings that had established after the 1994 cutting (i.e., about 8 years old), and without indication of browsing, were randomly selected in each plot, with the exception of one plot whereby only 12 saplings of similar size could be found. Four contrasting vegetation control treatments were compared (i.e., total circular removal [TC], total semicircular removal [TS], partial circular removal [PC], and no vegetation control [NC]). TC and TS treatments consisted, respectively, of manually removing all living vegetation using pruning shears around the entire perimeter and only removing vegetation in a 180° portion (facing south) of the crown of the target sapling. Using the same vegetation control technique, the PC treatment was applied by removing all the competing vegetation that reached three-quarters or more of the height of the target sapling. In all three treatments, vegetation was removed in a 0.75 m radius around the crown of the target sapling. All selected saplings in each plot received the same treatment. The four treatments were randomly assigned to the 32 plots ($n = 8$ per treatment).

Tree Measurements and Herbivory Estimations

The main stem height and diameter at 10 cm above the soil surface were measured on all selected saplings immediately after the application of the vegetation control treatments (June 2002) and remeasured in October 2003. Height increment (cm/yr) and diameter increment (mm/yr) were computed as the difference in height or diameter of saplings between the early spring 2002 (i.e., height and diameter in 2001) and the late fall 2003 (i.e., height and diameter in 2003) measurements divided by two. Relative height growth ($RHG = 0.5 (\text{Height}_{2003} - \text{Height}_{2001}) / \text{Height}_{2001}$), relative diameter growth ($RDG = 0.5 (\text{Diameter}_{2003} - \text{Diameter}_{2001}) / \text{Diameter}_{2001}$), and an etiolation index (RHG/RDG ratio) were then calculated for each sapling. The computation of an etiolation index (RHG/RDG ratio) rather than etiolation (height/diameter ratio) was preferred because it took into account the initial height and diameter of saplings. This index thus described the relative importance of biomass allocation into height versus diameter growth after treatment application. For similar heights, a higher etiolation index refers to an allocation to height growth to the detriment of diameter growth. For data analysis, all computed parameters were averaged at the plot scale (i.e., mean of all saplings of a plot). The number of dead saplings in the plots was also noted and used to compute survival

percentage as a function of vegetation control treatments. Finally, per-plot browsing proportion was assessed by calculating the proportion of saplings that presented visible browsing (i.e., loss of lateral and/or terminal tissues) within each plot. This measurement thus refers to a browsing occurrence rather than a browsing intensity per se.

Data Analyses

Analyses of variance for completely randomized designs were used to test for the effect of vegetation control treatments on the plot-averaged survival (after “arcsine ($x^{0.5}$)” transformation), height and diameter growth (increment and relative growth rates), browsing proportion (after “arcsine ($x^{0.5}$)” transformation), and mean sapling etiolation index of *B. alleghaniensis* saplings. Treatment averages were ranked using Tukey–Kramer post hoc test. Then, to better understand the effect of browsing occurrence on sapling height and diameter growth, as well as on sapling etiolation index, linear regression was performed, with per-plot browsing proportion as the causal variable and growth parameters as the response variable. This analysis was carried out separately for each vegetation control treatment. Also, to assess the importance of an interaction between browsing and vegetation control while simultaneously looking at the impact of the vegetation control treatment levels and the impact of browsing within groups, a test for homogeneity of slope (basically an analysis of covariance where per-plot browsing proportion and the interaction term are set as the covariates) was performed. Significance level was set at $\alpha = 0.05$ in all analyses. All analyses were performed using NCSS software (Hintze 2002).

Results

Initial Sapling Conditions

In sapling measurements before application of vegetation control treatments, there was no significant difference of

height and diameter ($F = 2.05$, $p = 0.1302$ and $F = 2.92$, $p = 0.0514$, respectively). There was a significant difference for etiolation ($F = 17.52$, $p = 0.0070$) between PC and NC, however (Table 1). To deal with these initial sapling etiolation differences, etiolation after two growing seasons was compared using relative height and diameter growth rates (i.e., RHG/RDG), referred to as the etiolation index, rather than final sapling height and diameter. Similarly, RHG and RDG were presented in addition to mean height or diameter increments to ensure consistency of conclusions.

Sapling Survival and Growth

Survival was very high in all vegetation control treatments, despite a significantly lower sapling survival in the TS compared to the TC treatment (Table 2; Fig. 1A). On average, survival was about 98.7%, and no mortality was recorded in the TC treatment. Mean diameter increment significantly increased with increasing vegetation control (Fig. 1B; Table 2), ranging from 2.99 mm/yr in the NC to 5.42 mm/yr in the TC. However, the NC treatment was only significantly different from the TC and PC treatments. There was no significant difference between treatments for mean height increment (Table 2; Fig. 1C). Mean height increment ranged from about 38 cm in the TC to 42 cm in the TS. RHG and RDG treatment response trends did not differ from those seen for mean height and diameter increments (data not shown, see statistics in Table 2). Mean etiolation index decreased significantly with increasing vegetation control, though TC and PC treatments were not significantly different (Table 2; Fig. 1D).

Browsing Effect and Interaction With Vegetation Control Treatments

Per-plot browsing proportion varied greatly within each vegetation control treatment, especially in the TC and PC treatments (from 0 to 32%). As a result, plot browsing proportion did not significantly differ among vegetation

Table 1. Mean sapling height, diameter, and etiolation before and two years after application of vegetation control treatments.

<i>Betula alleghaniensis</i> Saplings				
<i>Vegetation Control Treatment</i>	<i>Plots (n)</i>	<i>Height (cm)</i>	<i>Diameter (mm)</i>	<i>Etiolation (cm/mm)</i>
Before vegetation control treatments				
TC	8	162.23 (5.92)a	12.92 (0.64)a	13.47 (0.48)ab
PC	8	149.61 (7.88)a	12.12 (0.75)a	12.93 (0.75)a
TS	8	138.39 (4.44)a	10.51 (0.45)a	13.56 (0.45)ab
NC	8	148.74 (8.33)a	10.73 (0.80)a	14.68 (0.80)b
After vegetation control treatments				
TC	8	236.79 (6.22)a	23.77 (0.79)c	10.00 (0.28)a
PC	8	233.88 (9.23)a	21.53 (1.17)bc	10.92 (0.23)ab
TS	8	223.87 (3.88)a	18.60 (0.60)ab	12.09 (0.32)b
NC	8	225.20 (8.27)a	16.73 (0.95)a	13.60 (0.44)c

Vegetation control treatments were TC, PC, TS, and NC. Standard errors of means in parentheses. Contrasting letters correspond to significant differences between treatment means at $\alpha = 0.05$.

Table 2. Analysis of variance results for differences in survival, mean diameter and height growth, and etiolation index as a function of four contrasting vegetation control treatments (see Table 1 for treatment details).

Dependent Variable	Sources of Variation	df	Sum of Squares	F	p
Survival	Vegetation control treatment	3	0.19	4.00	0.0173
	Error	28	0.44		
Mean diameter increment	Vegetation control treatment	3	102.13	14.17	<0.0001
	Error	28	67.29		
Mean height increment	Vegetation control treatment	3	721.65	1.94	0.1468
	Error	28	3480.16		
RDG	Vegetation control treatment	3	0.06	4.06	<0.0001
	Error	28	0.15		
RHG	Vegetation control treatment	3	2.27	2.21	0.1093
	Error	28	0.10		
Mean etiolation index	Vegetation control treatment	3	261.47	17.52	<0.0001
	Error	28	139.32		

control treatments at $\alpha = 0.05$ but did differ at $\alpha = 0.1$ ($F = 2.63$, $p = 0.07$; Fig. 2). There was a trend of increased browsing with increasing intensity of vegetation removal (Fig. 2). A preliminary analysis (data not shown) indicated that browsing was independent of plot positions within the study site, plot orientation, or plot slope.

Variations in per-plot browsing proportion had relatively little effect on mean sapling growth and etiolation index, except in TC and TS treatments (Table 3; Fig. 3). Diameter growth was negatively correlated to per-plot browsing proportion in the most severe vegetation control treatments TC and PC, but the relationship was not significant for PC (Table 3). In contrast, this association was

positive in the less severe vegetation control treatment TS and NC, though the correlation was weak and only significant for the TS treatment. Height growth followed a similar trend in association with per-plot browsing proportion, although the relationship was significant only in the TS treatment (Table 3). Severe vegetation control treatments (TC and PC) also produced an opposite association of mean etiolation index with per-plot browsing proportion compared to TS and NC treatments, but the relationship was significant for TC only (Table 3).

Finally, analysis of homogeneity of slopes revealed differences in mean diameter increment and mean etiolation index between vegetation control treatments, especially at

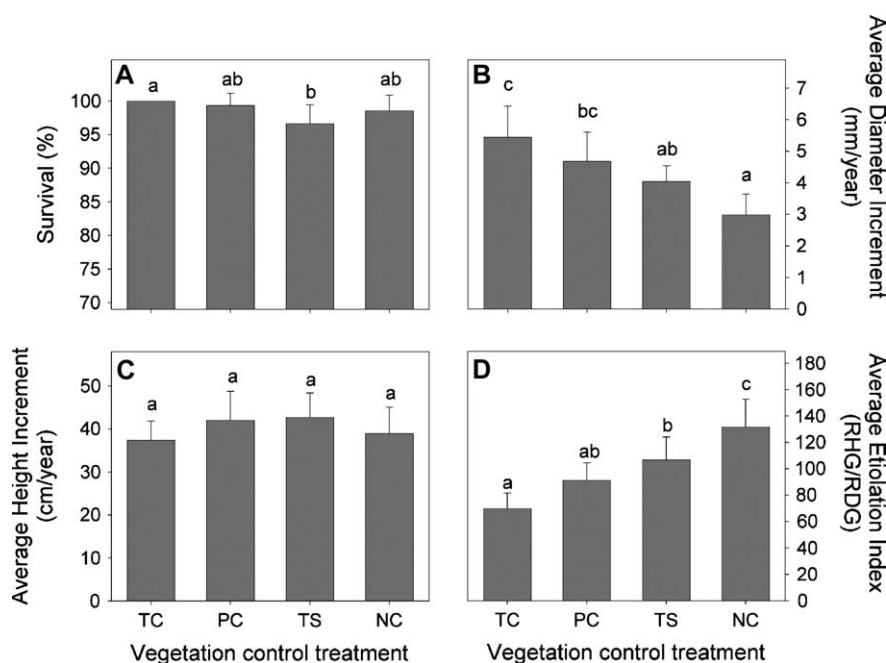


Figure 1. Mean and standard error for (A) survival, (B) mean sapling diameter growth, (C) mean sapling height growth, and (D) mean etiolation index as a function of four contrasting vegetation control treatments (see Table 1 for treatment details). Contrasting letters correspond to significant differences at $\alpha = 0.05$.

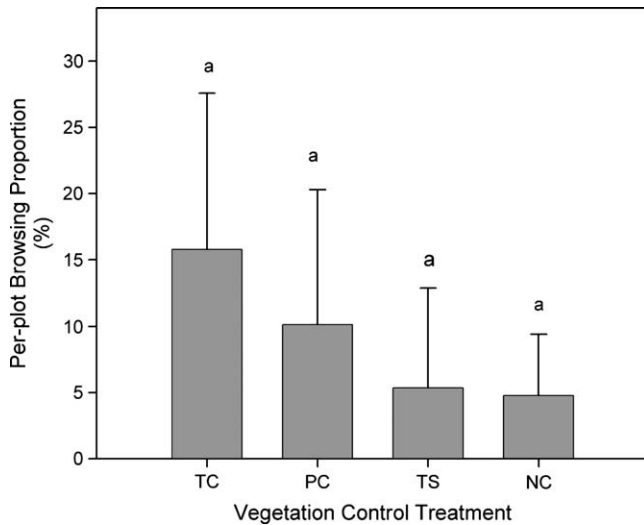


Figure 2. Mean and standard error for plot browsing proportion (%) as a function of four contrasting vegetation control treatments (see Table 1 for treatment details). Contrasting letters correspond to significant differences at $\alpha = 0.05$.

low proportions of per-plot browsing, although they decreased with increasing proportions of per-plot browsing (Fig. 3; Table 4).

Discussion

Sapling Response to Vegetation Control Treatment

Responses of eight-year-old saplings to applications of contrasting vegetation control treatments were closely related to the severity of competition removal imposed

Table 3. Summary of the linear regression parameters for the relationships between per-plot browsing proportion and sapling growth (RDG and RHG) and sapling mean etiolation index.

	Linear Regression Parameters			
	Intercept	Slope	r^2	p
RDG				
TC	0.4969	-0.6712	0.6589	0.0144
PC	0.3971	-0.3088	0.2124	0.2504
TS	0.3220	0.6333	0.7929	0.0030
NC	0.2374	0.6865	0.2125	0.2503
RHG				
TC	0.2502	-0.1005	0.0951	0.4574
PC	0.2886	0.0064	0.0001	0.9819
TS	0.2687	0.6348	0.6785	0.0119
NC	0.2313	0.5722	0.1597	0.3267
Mean etiolation index				
TC	0.5164	1.3063	0.6116	0.0218
PC	0.7972	1.2196	0.2003	0.2662
TS	1.0252	-1.2101	0.0965	0.4540
NC	1.1829	-0.5161	0.1375	0.3658

Results are presented for each vegetation control treatment (see Table 1 for treatment details).

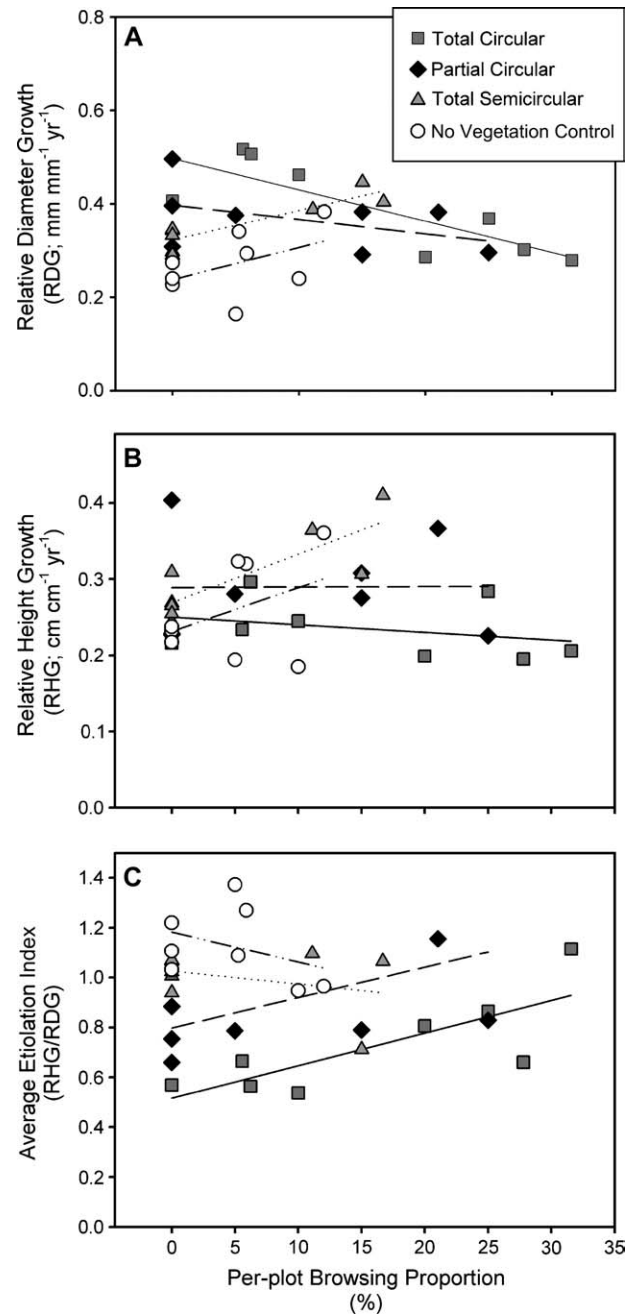


Figure 3. Relationship between per-plot browsing and (A) RDG; $\text{mm mm}^{-1} \text{yr}^{-1}$, (B) RHG; $\text{cm cm}^{-1} \text{yr}^{-1}$, and (C) mean etiolation index (RHG/RDG) for four contrasting vegetation control treatments (see Table 1 for treatment details).

by treatments (Balandier et al. 2006). Among treatments, significant increases in mean diameter increment were reported for the two most severe treatments TC and PC when compared to NC treatment. However, after two growing seasons, no significant improvement of height growth was detected in response to competition removal. With or without application of vegetation control treatments, *Betula alleghaniensis* saplings were able to

Table 4. Analysis of covariance results for the relative diameter (RDG) and height (RHG) growth and mean etiolation index.

Dependent Variable	Sources of Variation	df	Sum of Squares	F	p
RDG	Vegetation control	3	0.115	11.696	<0.0001
	Browsing	1	0.001	0.304	0.5865
	Interaction (main factor $\times$ covariate)	3	0.062	6.277	0.0027
	Error	24	0.078		
RHG	Vegetation control	3	0.007	0.718	0.5507
	Browsing	1	0.011	3.459	0.0752
	Interaction (main factor $\times$ covariate)	3	0.019	2.081	0.1294
	Error	24	0.074		
Mean etiolation index	Vegetation control	3	1.080	16.776	<0.0001
	Browsing	1	0.002	0.106	0.7475
	Interaction (main factor $\times$ covariate)	3	0.202	3.144	0.0437
	Error	24	0.515		

In the analysis, vegetation control (see Table 1 for treatment details) was set as the main factor and per-plot browsing proportion as the covariate.

maintain their dominant or codominant status in regeneration and avoid being topped by competing vegetation. Etiolation clearly allowed saplings in the NC and, to some extent, in the TS treatments to exhibit similar height growth to saplings in more severe vegetation control treatments.

Betula alleghaniensis' ability to preferentially invest in height versus diameter growth has previously been reported as a mechanism of response to competition for light (Bellefleur & Laroque 1983; Delagrangue et al. 2004), which is critical in these types of forest (Ricard et al. 2003). However, maximizing height growth might be conducive to lower overall sapling vigor, especially over a prolonged period of time. Sustained etiolation results in lower stem resistance to possible breakage, and due to the competitive allocation of carbon in limited environments, etiolation could be associated with a reduction in lateral crown development, which in turn could decrease light interception efficiency (Givnish 1995; Beaudet & Messier 1998; Sterck 1999).

Our results generally confirm the conclusions of earlier studies that showed an improvement in diameter growth over height growth with vegetation control treatments such as thinning or surrounding vegetation removal for *B. alleghaniensis* and other tree species (e.g., Smith & Lamson 1983; Lamson 1988; Robitaille et al. 1990; Miller 2000; Kneeshaw et al. 2002). Kneeshaw et al. (2002) explained that suppressed established coniferous individuals of all size classes first responded with an increase in root and radial growth after release treatment. According to the authors, height growth was delayed because of the need for previously suppressed individuals to modify their root/shoot ratio to counterbalance the higher water consumption they face.

Our study differs mainly from others in that we tested different types of vegetation control treatments using a manual removal method that corresponded to different intensities of vegetation control. Results, indicate, however, as do most other studies on facilitation of *B. alleghaniensis* regeneration, that despite marginal differences in sapling height growth two years after treatments, application of vegetation control, especially TC and PC treat-

ments, was generally superior to the absence of treatment in decreasing sapling signs of stress. Another advantage of severe vegetation control is reducing environmental variability or heterogeneity of the stand, which may reduce future mortality (Nilsson & Allen 2003; see also the review by Balandier et al. 2006).

Interaction Between Vegetation Control Treatment and Browsing Proportion

Although the population of Virginia deer (*Odocoileus virginianus*) was considered low to moderate at our study site (2–3 individuals/km²) and the proportion of browsed saplings in plots never reached more than 32.5%, browsing increased with the intensity of vegetation removal (especially in the TC treatment), which made saplings more conspicuous (Kohn & Mooty 1971; Drolet 1978; Monthey 1984). The increase in browsing occurrence with intensity of vegetation removal was not directly proportional among all treatments. Indeed, when applying TC or PC vegetation control treatments, browsing doubled on average compared to the TS or NC treatment. This tended to create two distinct systems discriminated by the type of relationship reported between browsing and mean growth rates. Diameter and height growth in the TS and NC treatments were relatively independent of browsing proportion, whereas negative relationships were reported between growth rates and browsing proportion in the TC and PC treatments. Because of a greater visibility of saplings in the latter treatments, diameter growth increments are mitigated by an increase in the occurrence and possibly the severity of damage caused by deer. Furthermore, increasing resource availability (i.e., light and nutrients) owing to more severe competition removal may increase the nutritional value of tissues, which has been demonstrated to favor palatability to herbivores (Hughes & Fahey 1991; Dumont et al. 2005). This may be particularly true because *B. alleghaniensis* is a desirable forage species for herbivores (Pearce 1937; Hannah 1978; Hughes & Fahey 1991).

Unfortunately, it is difficult to report conclusively on the possible effects of type of browsing and level of severity in circular treatments because relative intensity and localization in the crown were not recorded. However, it is reasonable to assume, because of contrasting arrangements of competing vegetation around the released saplings, that similar browsing occurrence in plots may reflect dissimilar browsing severities and impacts on growth. For example, in the TS treatment, half of the crown is intermingled with surrounding vegetation, protecting half of the sapling crown and exposing the other half primarily to lateral browsing. In contrast, circular treatments (i.e., TC and PC) offer localization and accessibility to the whole-crown circumference at a lower foraging cost, which are key factors in deer winter foraging strategy (Dumont et al. 2005).

Thus, even if more research must be done concerning the assessment of actual impacts of contrasting types of browsing (e.g., terminal or lateral) on sapling growth and vigor, browsing occurrence was damaging enough in TC and PC treatments to invalidate growth enhanced by competition removal. This is not surprising because browsing is recognized as seriously affecting crown development and overall individual vigor (e.g., Bulinski & McArthur 1999; Moore et al. 1999; de la Cretaz & Kelty 2002; Di Stefano 2005). In the case of high herbivore densities, consequences of browsing can also lead to decreases in survival and shifts in future stand composition (Facelli 1994; de la Cretaz & Kelty 2002; Bellingham & Allan 2003). This leads to the conclusion that particular care must be taken in using vegetation control treatments depending on local herbivore densities.

Conclusions

Control of competing vegetation is recommended to ensure sufficient *Betula alleghaniensis* crop trees within juvenile stands that do not have adequate sapling densities (Nolet et al. 2001a). Our results clearly show that application of vegetation control treatments increased growth (especially in diameter) and vigor of *B. alleghaniensis* saplings, more so as vegetation removal intensity increased. Because browsing risk increases with vegetation removal intensity and damage severity is generally linked with herbivore density (Bulinski & McArthur 2003), the decision to apply a vegetation control treatment must also consider local herbivore population densities. In our particular case, the low to moderate density of Virginia deer was sufficient to cause enough browsing in some plots of the TC and PC treatments to counter the vigor and diameter growth enhancements induced by these treatments.

While a specific study on the economic costs of the contrasting treatments is needed, we observed that the TS treatment required slightly more than halftime it took to apply the TC or PC treatment, which suggests

a much higher costs for the latter. Taking into consideration the growth enhancement and browsing occurrence observed in the TC and PC treatments, we suggest that these treatments should be applied when large mammal herbivore densities are low to moderate. Such treatments will promote better vigor and likely generate higher rates of success in developing dominant *B. alleghaniensis* canopy trees. Furthermore, considering the small differences in growth gain between the TC and the PC treatments, the latter should be considered as the treatment of choice because it might favor future stem quality by (1) limiting the growth of new lateral branches and (2) maintaining the natural dynamic of pruning of lower branches. In the case of higher herbivore population densities in the regenerating area, less severe and less time-consuming vegetation removal treatments are preferred in light of the trade-off between lowered vigor due to competition and lowered risk of browsing. For these reasons, the TS treatment presents several advantages such as keeping browsing very low while greatly reducing competition. Moreover, it would be interesting to examine how orientation of the halfcircle of the TS treatment—say in alternative direction to protect at least 50% of target saplings or in opposite direction to known cervidae paths—may affect the trade-off between vigor and herbivory. Much more insight might be gained if a similar study could be conducted at a larger scale to include a wide range of herbivore densities.

Implications for Practice

- Increasing intensity of vegetation control increases growth (mainly in diameter) and vigor of *Betula alleghaniensis* saplings established in juvenile stands, especially when browsing occurrence is low.
- The choice in the intensity of vegetation removal around a target sapling must also include consideration of the local herbivore population densities, because differences between vegetation control treatments decrease with increasing browsing occurrence.
- With respect to treatment operation costs and sapling stem quality considerations, semicircular (TS) competition removal or even no vegetation control should be preferred where deer population density is low to moderate (2–3 individuals/km²), whereas PC competition removal is recommended where deer populations are very low to absent.

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